

FDU 数值算法 5. 微分方程数值解法

本文根据邵美悦老师课堂笔记整理而成，并参考以下教材：

- 微分方程数值解 (陈文斌 & 程晋 & 吴新明 & 李立康) 第 2, 3, 4, 5 章
- 数值分析 (李庆扬 & 王能超 & 易大义, 第 5 版) 第 9 章
- 数值计算方法 (吕同富 & 康兆敏 & 方秀男) 第 9 章

欢迎批评指正!

5.1 An Introduction

5.1.1 常微分方程初值问题

一阶常微分方程初值问题的标准形式如下：

$$\begin{aligned}\frac{du(t)}{dt} &= f(t, u(t)) \\ u(t_0) &= u_0\end{aligned}$$

以一阶线性时不变系统为例：

$$\begin{aligned}\frac{du(t)}{dt} &= Au(t) + b(t) \\ u(t_0) &= u_0\end{aligned}$$

where $A \in \mathbb{C}^{n \times n}$ and $u(t), b(t) \in \mathbb{C}^n$

其解为：

$$u(t) = e^{A(t-t_0)}u_0 + \int_{t_0}^t e^{A(t-s)}b(s)ds$$

- 特殊地，考虑 $n = 1$ 的齐次情况：(设 $a \in \mathbb{R}$)

$$\begin{aligned}\frac{du(t)}{dt} &= au(t) \\ u(t_0) &= u_0\end{aligned}$$

其解为 $u(t) = u_0 e^{a(t-t_0)}$

当 $a > 0$ 时， $u(t)$ 随 t 指数增长趋于无穷大 (具体是正无穷还是负无穷取决于 u_0 的符号)

当 $a < 0$ 时， $u(t)$ 随 t 指数衰减趋于 0 (若 $a \in \mathbb{C}$ ，则此条件变为 $\text{Re}(a) < 0$)

当 $a = 0$ 时， $u(t) \equiv u_0$ 为常数函数。

- 考虑 $n = 1$ 的非齐次情况：

$$\begin{aligned}\frac{du(t)}{dt} &= au(t) + b(t) \\ u(t_0) &= u_0\end{aligned}$$

其解为：

$$\begin{aligned}u(t) &= e^{a(t-t_0)} \left[u_0 + \int_{t_0}^t b(s)e^{-a(s-t_0)}ds \right] \\ &= u_0 e^{a(t-t_0)} + \int_{t_0}^t b(s)e^{a(t-s)}ds\end{aligned}$$

- 更一般的, 考虑 $n = 1$ 的时变情况:

$$\begin{aligned}\frac{du(t)}{dt} &= a(t)u(t) + b(t) \\ u(t_0) &= u_0\end{aligned}$$

其解为:

$$u(t) = \exp\left(\int_{t_0}^t a(s)ds\right) \left[u_0 + \int_{t_0}^t b(s) \exp\left(-\int_{t_0}^s a(\tau)d\tau\right)ds\right]$$

(微分方程数值解, 定理 2.1.5)

若 $f(t, x)$ 在区域 $\mathcal{D} := \{(t, x) : |t - t_0| \leq a, \|x - x_0\| \leq b\}$ 上是连续的, 且关于 x 满足 Lipschitz 条件, 即存在实数 $L > 0$ 使得:

$$|f(t, x_1) - f(t, x_2)| \leq L|x_1 - x_2| \quad (\forall (t, x_1), (t, x_2) \in \mathcal{D})$$

则一阶常微分方程初值问题至少在区间 $I := \{t : |t - t_0| \leq h\}$ 上存在唯一解, 其中 $h = \min\{a, b/M\}$, M 为 $|f(t, x)|$ 在区域 $\mathcal{D}$ 上的最大值.

- (存疑: L 与 t 有关?)
- 若将 f 的条件加强, 例如 $f(t, x)$ 在 $\mathcal{D}$ 上导数一致有界, 则区间 I 可延拓到 $[a, b]$.
若将 f 的条件减弱, 例如把 f 关于 x 的 Lipschitz 条件去掉, 则存在性还成立, 但唯一性不能保证.

考虑一阶常微分方程初值问题:

$$\begin{aligned}\frac{du(t)}{dt} &= f(t, u(t)) \\ u(t_0) &= u_0\end{aligned}$$

其数值解法, 就是求 $u(t)$ 在一系列离散节点 $t_0, t_1, \dots, t_n, \dots$ 上的近似值 $u_0, u_1, \dots, u_n, \dots$. 我们默认节点等距, 间距记为 h , 即 $t_n = t_0 + nh$ ($\forall n \in \mathbb{N}$)

求解过程按节点排列的次序一步步向前推进.

若计算 u_{n+1} 时只用到前一点 u_n 的值, 则称为**单步法**,

若计算 u_{n+1} 时用到前面多个点的值, 则称为**多步法**.

单步法是多步法的基础, 因为多步法最开始的几个近似值需要通过单步法得到.

5.1.2 差商逼近法

差商逼近法就是用差商逼近导数值, 从而将初值问题离散化的方法.

例如可取:

$$u'(t_n) \approx \frac{u(t_{n+1}) - u(t_n)}{h} = \frac{u(t_n) - u(t_{n+1})}{-h} \approx u'(t_{n+1})$$

引入参数 $\alpha \in [0, 1]$, 考虑凸组合:

$$\alpha u'(t_n) + (1 - \alpha) u'(t_{n+1}) \approx \frac{u(t_{n+1}) - u(t_n)}{h}$$

于是我们有:

$$\begin{aligned}
u(t_{n+1}) &\approx u(t_n) + h[\alpha u'(t_n) + (1 - \alpha)u'(t_{n+1})] \\
&= u(t_n) + h[\alpha f(t_n, u(t_n)) + (1 - \alpha)f(t_{n+1}, u(t_{n+1}))] \\
&\quad \Updownarrow \\
u_{n+1} &\approx u_n + h[\alpha f_n + (1 - \alpha)f_{n+1}]
\end{aligned}$$

- 取 $\alpha = 1$ 即得**显式 Euler 法**:

$$u_{n+1} = u_n + hf_n$$

- 取 $\alpha = 0$ 即得**隐式 Euler 法**:

$$u_{n+1} = u_n + hf_{n+1}$$

- 取 $\alpha = \frac{1}{2}$ 即得**隐式梯形法**:

$$u_{n+1} = u_n + \frac{h}{2}(f_n + f_{n+1})$$

近似替换 $f_{n+1} = f(t_{n+1}, u_{n+1}) \approx f(t_{n+1}, u_n + hf(t_n, u_n)) = f(t_{n+1}, u_n + hf_n)$ 即得**显式梯形法**:

$$u_{n+1} = u_n + \frac{h}{2}(f_n + f(t_{n+1}, u_n + hf_n))$$

5.1.3 Taylor 展开法

考虑将常微分方程的精确解 $u(t)$ 在 t_n 时刻 Taylor 展开, 那么有:

$$u(t_{n+1}) = u(t_n) + hu'(t_n) + \frac{h^2}{2!}u''(t_n) + \cdots + \frac{h^k}{k!}u^{(k)}(t_n) + O(h^{k+1})$$

其中:

$$\begin{aligned}
u'(t_n) &= f(t_n, u(t_n)) \\
u''(t_n) &= \frac{df}{dt} = f'_t + f'_u f \\
u'''(t_n) &= \frac{d^2f}{dt^2} = f''_{tt} + 2f''_{tu}f + f''_{uu}f^2 + f'_u(f'_t + f'_u f)
\end{aligned}$$

其中下标 t, u 表示相应的偏导数.

我们可用待定系数法结合 Taylor 展开推导 Runge-Kutta 类方法.

5.1.4 数值积分法

数值积分法的基本思想是将微分方程转化为积分方程:

(我们以前说过, 数值积分的性质要明显优于数值微分)

$$\begin{aligned}
\frac{du(t)}{dt} &= f(t, u(t)) \\
u(t_0) &= u_0 \\
&\quad \Updownarrow \\
u(t_m) - u(t_n) &= \int_{t_n}^{t_m} f(t, u(t))dt \quad (m > n) \\
u(t_0) &= u_0
\end{aligned}$$

我们可对上述积分方程应用数值积分方法离散化，从而获得原初值问题的一个离散差分格式。
例如取 $m = n + 1$ ，并应用矩形求解公式，即得：

$$u_{n+1} = u_0 + hf(\theta t_n + (1 - \theta)t_{n+1}, \theta u_n + (1 - \theta)u_{n+1}) \quad (\text{for some } \theta \in [0, 1])$$

常见积分方程：

- ① 第一类 Fredholm 积分方程：

$$-\int_0^1 \kappa(x, y)f(y)dy = b(x)$$

- ② 第二类 Fredholm 积分方程：

$$f(x) - \int_0^1 \kappa(x, y)f(y)dy = b(x)$$

其中 $f(\cdot)$ 的性质未被积分完全吃掉，性质比第一类 Fredholm 积分方程更好。

- ③ Volterra 积分方程：

$$f(x) - \int_0^x \kappa(x, y)f(y)dy = b(x)$$

令 $\tilde{\kappa}(x, y) = \begin{cases} \kappa(x, y), & \text{if } 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$ 就变为第二类 Fredholm 积分方程。

尽管数学上 Volterra 积分方程隶属于第二类 Fredholm 积分方程，
但求解时不一定按第二类 Fredholm 积分方程的方法解。

5.2 单步法

5.2.1 Euler 法

考虑一阶常微分方程初值问题：

$$\begin{aligned} \frac{du(t)}{dt} &= f(t, u(t)) \\ u(t_0) &= u_0 \end{aligned}$$

其显式 Euler 迭代格式为：

$$\begin{aligned} &\text{given } u_0 = u(t_0) \\ &u_{n+1} = u_n + hf_n \\ &\text{where } t_n := t_0 + nh, u_n := u(t_n) \text{ and } f_n := f(t_n, u(t_n)) = f(t_n, u_n) \end{aligned}$$

单步截断误差为 $O(h^2)$ ，全局截断误差为 $O(h)$

其隐式 Euler 迭代格式为：

$$\begin{aligned} &\text{given } u_0 = u(t_0) \\ &u_{n+1} = u_n + hf_{n+1} \\ &\text{where } t_n := t_0 + nh, u_n := u(t_n) \text{ and } f_n := f(t_n, u(t_n)) = f(t_n, u_n) \end{aligned}$$

其中 u_{n+1} 是通过数值求解方程 $u_{n+1} = u_n + hf_{n+1} = u_n + hf(t_{n+1}, u_{n+1})$ 得到的，
计算量更大，但数值稳定性更高，不过单步截断误差仍为 $O(h^2)$ ，全局截断误差为 $O(h)$

5.2.2 梯形法

考虑一阶常微分方程初值问题:

$$\begin{aligned}\frac{du(t)}{dt} &= f(t, u(t)) \\ u(t_0) &= u_0\end{aligned}$$

其显式梯形迭代格式为:

$$\begin{aligned}\text{given } u_0 &= u(t_0) \\ u_{n+1} &= u_n + \frac{h}{2}(f_n + f(t_{n+1}, u_n + hf_n)) \\ \text{where } t_n &:= t_0 + nh, u_n := u(t_n) \text{ and } f_n := f(t_n, u(t_n)) = f(t_n, u_n)\end{aligned}$$

单步截断误差为 $O(h^3)$, 全局截断误差为 $O(h^2)$

其隐式梯形迭代格式为:

$$\begin{aligned}\text{given } u_0 &= u(t_0) \\ u_{n+1} &= u_n + \frac{h}{2}(f_n + f_{n+1}) \\ \text{where } t_n &:= t_0 + nh, u_n := u(t_n) \text{ and } f_n := f(t_n, u(t_n)) = f(t_n, u_n)\end{aligned}$$

其中 u_{n+1} 是通过数值求解方程 $u_{n+1} = u_n + \frac{h}{2}(f_n + f_{n+1}) = u_n + \frac{h}{2}(f_n + f(t_{n+1}, u_{n+1}))$ 得到的,

计算量更大, 但数值稳定性更高, 不过单步截断误差仍为 $O(h^3)$, 全局截断误差为 $O(h^2)$

5.2.3 Runge-Kutta 法

Runge-Kutta 法通过对更多点的近似导数值进行线性组合来构造更高精度的迭代公式:

$$\begin{aligned}u_{n+1} &= u_n + h \sum_{j=1}^r b_j s_j \\ \text{where } s_i &= f\left(t_n + c_i h, u_n + h \sum_{j=1}^r a_{i,j} s_j\right) \quad (i = 1, \dots, r)\end{aligned}$$

其中 $c_k, b_k, a_{k,j}$ 均为常数 (满足 $\sum_{j=1}^r b_j = 1$), 对应的迭代格式称为 r 阶 Runge-Kutta 法.

考虑标准测试问题:

$$\begin{aligned}\text{Given } \lambda &\in \mathbb{C} \\ u'(t) &= \lambda u(t) \quad (t > 0) \\ u(0) &= u_0 \neq 0\end{aligned}$$

其解为 $u(t) = u_0 e^{\lambda t}$, 它满足:

$$\lim_{t \rightarrow \infty} u(t) = 0 \quad \Leftrightarrow \quad \text{Re}(\lambda) < 0$$

Runge-Kutta 方法的迭代解具有形式 $u_k = (S(\lambda h))^k u_0$, 其中 $S(z)$ 是某个有理函数.

标准测试问题的绝对稳定域为 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$,

参数和步长的乘积 $z = \lambda h$ 需要落在 $\mathcal{A}$ 中, 该方法对标准测试问题才是稳定的.

若 $\{z \in \mathbb{C} : \text{Re}(z) < 0\} \subset \mathcal{A}$,

则只要 $\text{Re}(\lambda) < 0$, 无论步长 $h > 0$ 为多少, 该方法对标准测试问题都是稳定的.

定义 **Butcher 表** 为:

c_1	$a_{1,1}$	$a_{1,2}$	$\cdots$	$a_{1,r}$
c_2	$a_{2,1}$	$a_{2,2}$	$\cdots$	$a_{2,r}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
c_r	$a_{r,1}$	$a_{r,2}$	$\cdots$	$a_{r,r}$
	b_1	b_2	$\cdots$	b_r

- 严格下三角部分是对过去的依赖
- 对角线部分是对自身的依赖
- 严格上三角部分是对未来的依赖

若 A 是严格下三角阵, 则是显式迭代格式 (仅依赖于之前的状态), 可直接计算得到当前状态 u_{n+1} .

若 A 不是严格下三角阵, 则是隐式迭代格式 (不仅依赖于之前的状态, 还依赖于当前状态), 需要解非线性方程得到当前状态 u_{n+1} .

在此意义下, Euler 法和梯形法也属于 Runge-Kutta 法:

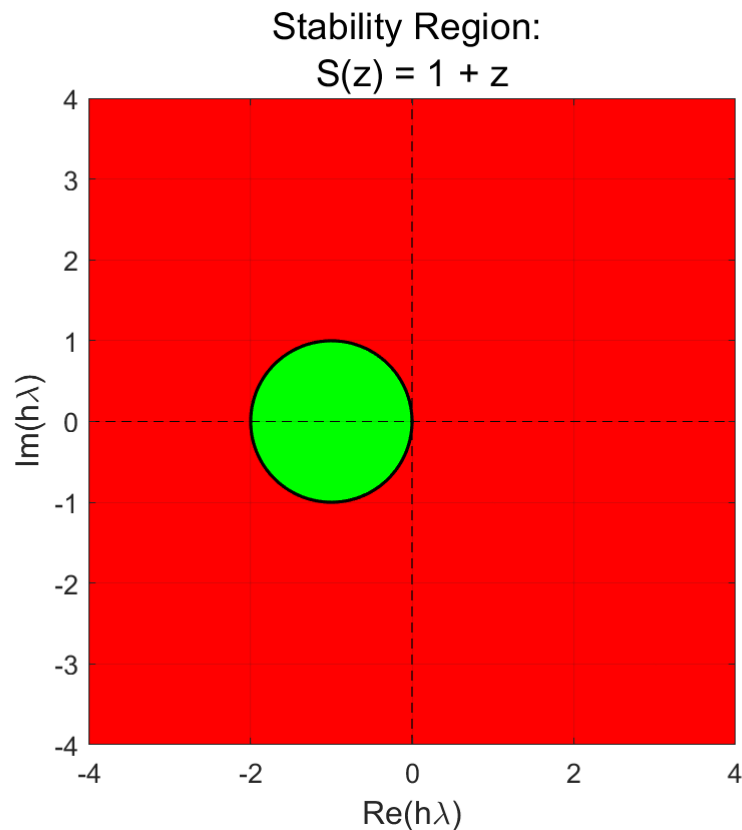
- **显式 Euler 法:**

$$\begin{aligned}
 u_{n+1} &= u_n + hf_n \\
 &\Updownarrow \\
 s_1 &= f_n = f(t_n, u_n) \\
 u_{n+1} &= u_n + hs_1 \\
 \hline
 \begin{array}{c|c} c_1 & a_{1,1} \\ \hline & b_1 \end{array} &= \begin{array}{c|c} 0 & 0 \\ \hline & 1 \end{array}
 \end{aligned}$$

单步截断误差为 $O(h^2)$, 全局截断误差为 $O(h)$

对于标准测试问题, $S(z) = 1 + z$

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:



- **隐式 Euler 法:**

$$\begin{aligned}
 u_{n+1} &= u_n + hf_{n+1} \\
 &\Updownarrow \\
 s_1 &= f_{n+1} = f(t_{n+1}, u_{n+1}) = f(t_n + h, u_n + hs_1) \\
 u_{n+1} &= u_n + hs_1
 \end{aligned}$$

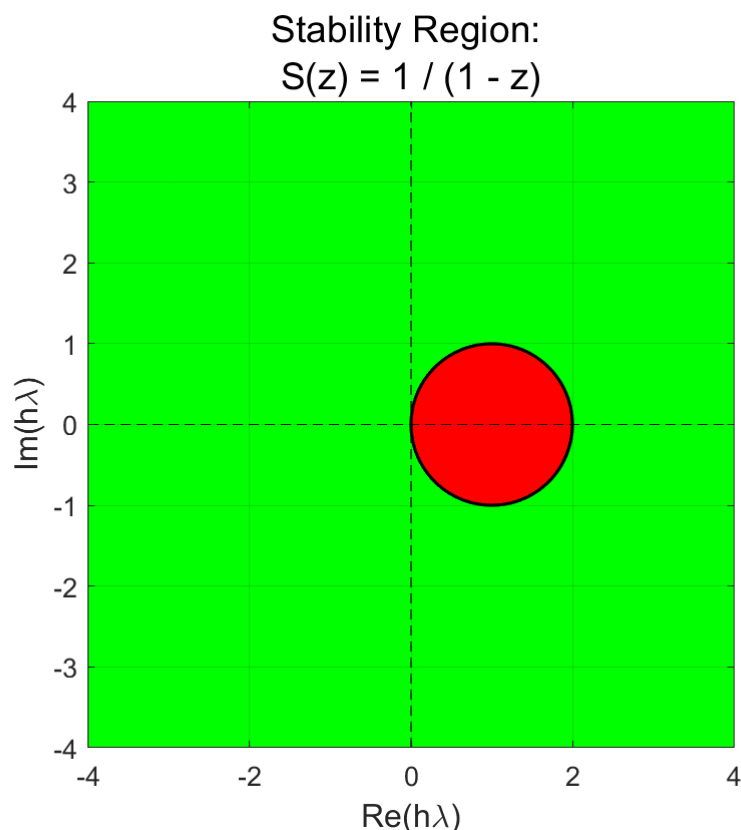
$$\begin{array}{c|c} c_1 & a_{1,1} \\ \hline & b_1 \end{array} = \begin{array}{c|c} 1 & 1 \\ \hline & 1 \end{array}$$

单步截断误差为 $O(h^2)$, 全局截断误差为 $O(h)$

对于标准测试问题, $S(z) = (1 - z)^{-1}$

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:

(可以看出隐格式的绝对稳定域更大, 比显格式更稳定)



由于 $\{z \in \mathbb{C} : \text{Re}(z) < 0\} \subset \mathcal{A}$, 故隐式 Euler 法对标准测试问题绝对稳定.

• **显式梯形法:**

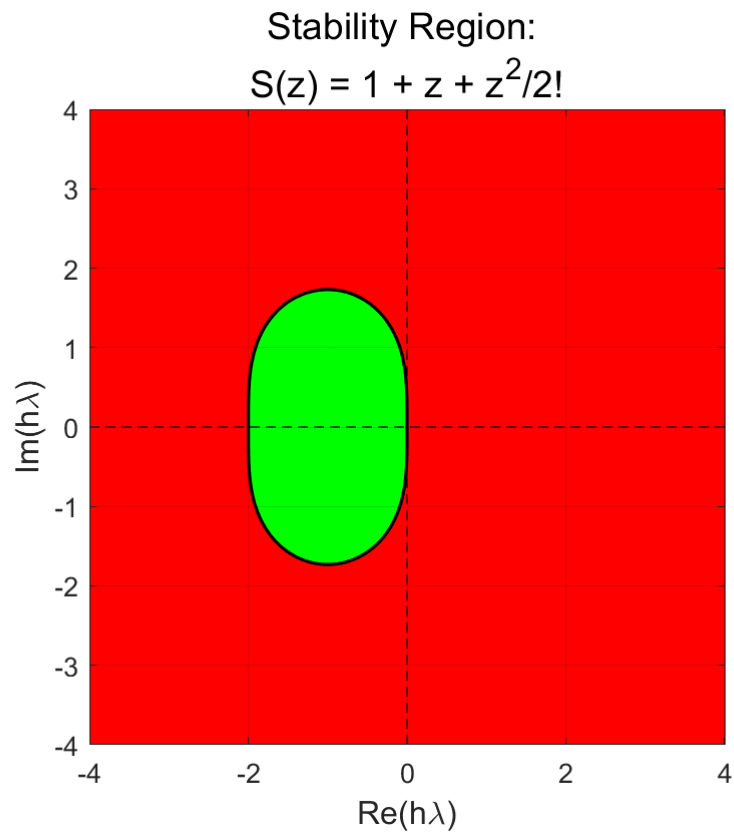
$$\begin{aligned}
 u_{n+1} &= u_n + \frac{h}{2}(f_n + f(t_{n+1}, u_n + hf_n)) \\
 &\Updownarrow \\
 s_1 &= f_n = f(t_n, u_n) \\
 s_2 &= f(t_{n+1}, u_n + hf_n) = f(t_{n+1}, u_n + hs_1) \\
 u_{n+1} &= u_n + \frac{h}{2}s_1 + \frac{h}{2}s_2
 \end{aligned}$$

$$\begin{array}{c|ccc|cc} c_1 & a_{1,1} & a_{1,2} & 0 & 0 & 0 \\ c_2 & a_{2,1} & a_{2,2} & 1 & 1 & 0 \\ \hline & b_1 & b_2 & & 1/2 & 1/2 \end{array}$$

单步截断误差为 $O(h^3)$, 全局截断误差为 $O(h^2)$

对于标准测试问题, $S(z) = 1 + z + z^2/2!$,

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:



- 隐式梯形法:

$$u_{n+1} = u_n + \frac{h}{2}(f_n + f_{n+1})$$

$\Updownarrow$

$$s_1 = f_n = f(t_n, u_n)$$

$$s_2 = f_{n+1} = f(t_{n+1}, u_{n+1}) = f\left(t_{n+1}, u_n + \frac{h}{2}s_1 + \frac{h}{2}s_2\right)$$

$$u_{n+1} = u_n + \frac{h}{2}s_1 + \frac{h}{2}s_2$$

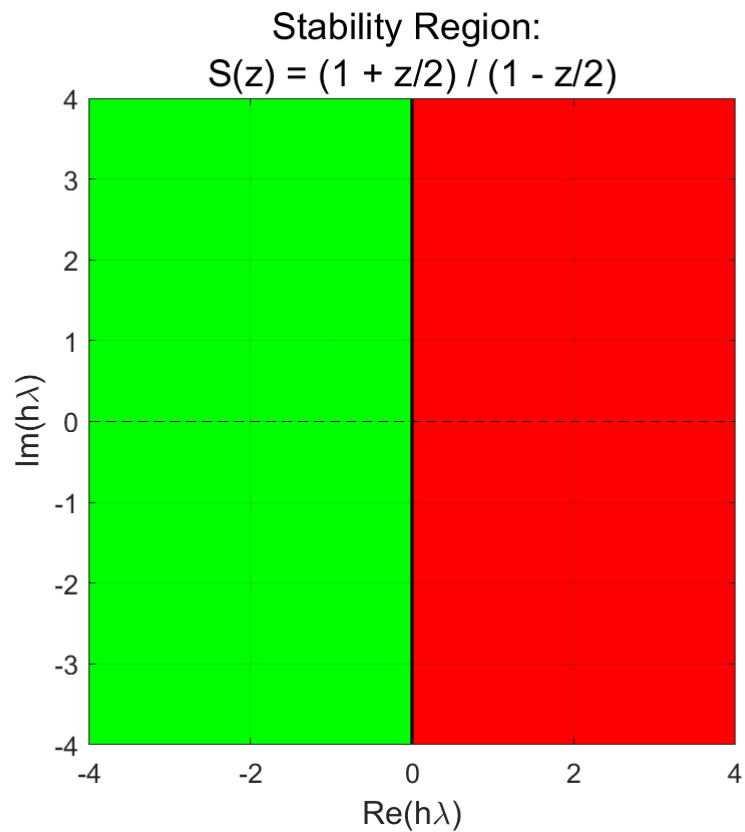
c_1	$a_{1,1}$	$a_{1,2}$	0	0	0
c_2	$a_{2,1}$	$a_{2,2}$	$= 1$	$1/2$	$1/2$
	b_1	b_2		$1/2$	$1/2$

单步截断误差为 $O(h^3)$, 全局截断误差为 $O(h^2)$

对于标准测试问题, $S(z) = (1 + z/2)(1 - z/2)^{-1}$

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:

(可以看出隐格式的绝对稳定域更大, 比显格式更稳定)



数值计算方法第 9.3 节给出:

- ① 二阶 Runge-Kutta 法: (标准中点法)

$$u_{n+1} = u_n + hf \left(t_n + \frac{h}{2}, u_n + \frac{h}{2} f(t_n, u_n) \right)$$

$\Updownarrow$

$$s_1 = f_n = f(t_n, u_n)$$

$$s_2 = f \left(t_n + \frac{h}{2}, u_n + \frac{h}{2} f(t_n, u_n) \right) = f \left(t_n + \frac{h}{2}, u_n + \frac{h}{2} s_1 \right)$$

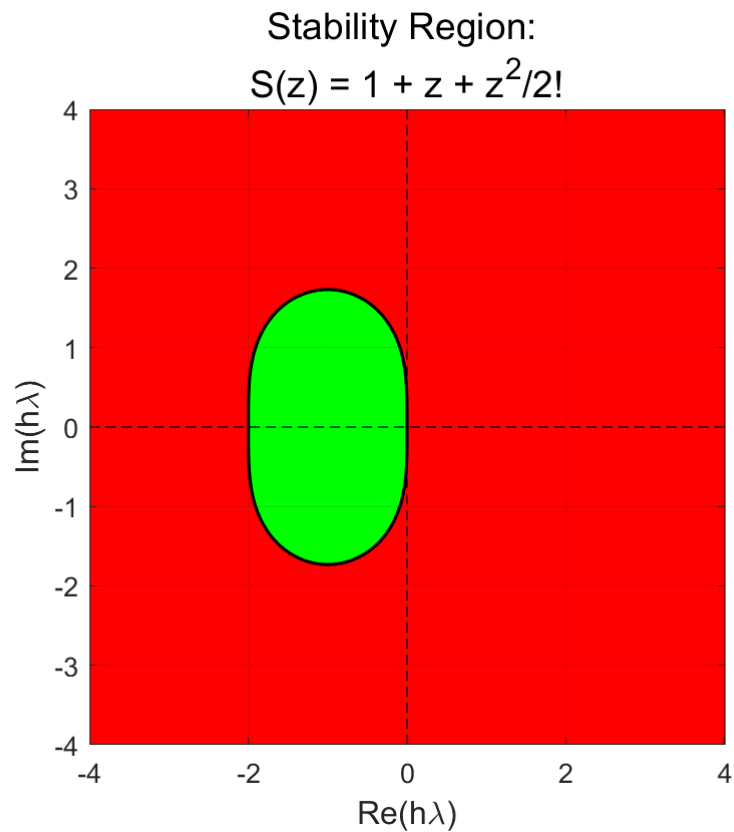
$$u_{n+1} = u_n + hs_2$$

c_1	$a_{1,1}$	$a_{1,2}$	0	0	0
c_2	$a_{2,1}$	$a_{2,2}$	$1/2$	$1/2$	0
	b_1	b_2		0	1

单步截断误差为 $O(h^3)$, 全局截断误差为 $O(h^2)$

对于标准测试问题, $S(z) = 1 + z + z^2/2!$

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:



• ② 三阶 Runge-Kutta 法:

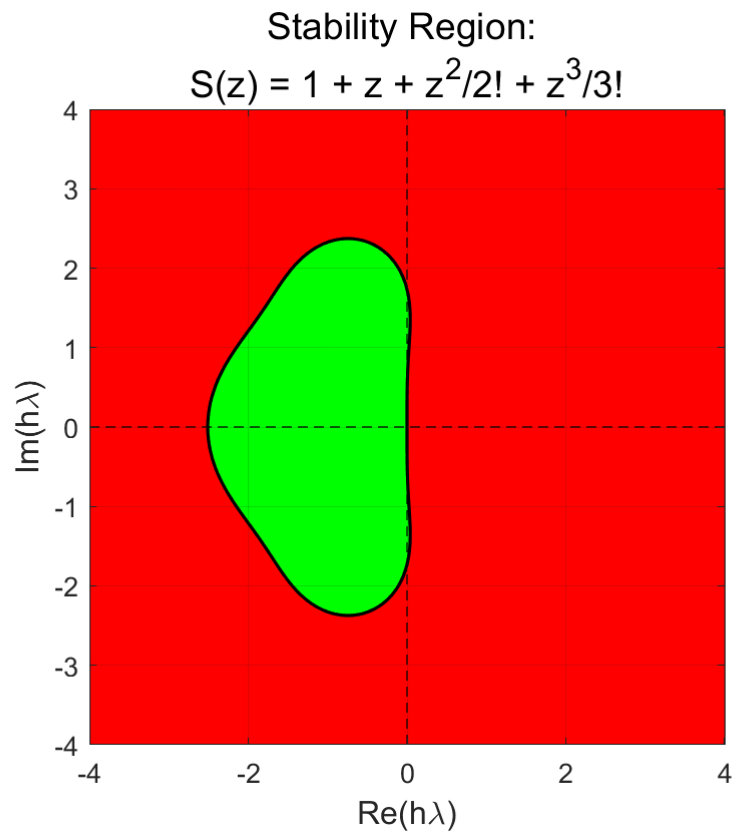
$$\begin{aligned}
 s_1 &= f_n = f(t_n, u_n) \\
 s_2 &= f\left(t_n + \frac{h}{2}, u_n + \frac{h}{2}s_1\right) \\
 s_3 &= f(t_n + h, u_n - hs_1 + 2hs_2) \\
 u_{n+1} &= u_n + \frac{h}{6}(s_1 + 4s_2 + s_3)
 \end{aligned}$$

c_1	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	0	0	0	0
c_2	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$	$1/2$	$1/2$	0	0
c_3	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$	1	-1	2	0
	b_1	b_2	b_3		$1/6$	$2/3$	$1/6$

单步截断误差为 $O(h^4)$, 全局截断误差为 $O(h^3)$

对于标准测试问题, $S(z) = 1 + z + z^2/2! + z^3/3!$

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:



• ③ 四阶 Runge-Kutta 法: (经典 RK4)

$$s_1 = f_n = f(t_n, u_n)$$

$$s_2 = f\left(t_n + \frac{h}{2}, u_n + \frac{h}{2}s_1\right)$$

$$s_3 = f\left(t_n + \frac{h}{2}, u_n + \frac{h}{2}s_2\right)$$

$$s_4 = f(t_n + h, u_n + hs_3)$$

$$u_{n+1} = u_n + \frac{h}{6}(s_1 + 2s_2 + 2s_3 + s_4)$$

c_1	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	$a_{1,4}$	0	0	0	0	0
c_2	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$	$a_{2,4}$	$1/2$	$1/2$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$	$a_{3,4}$	$1/2$	0	$1/2$	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	$a_{4,4}$	1	0	0	1	0
	b_1	b_2	b_3	b_4		$1/6$	$1/3$	$1/3$	$1/6$

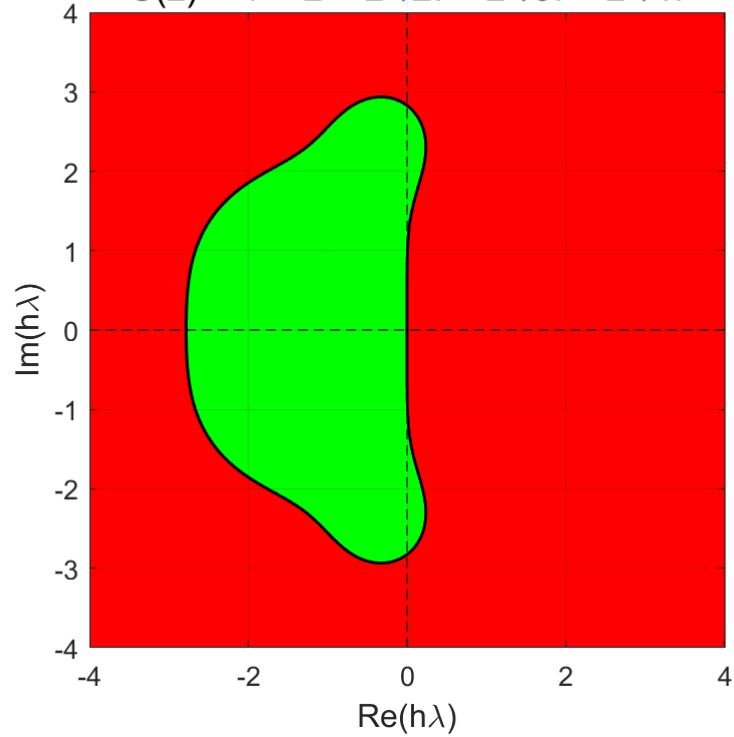
单步截断误差为 $O(h^5)$, 全局截断误差为 $O(h^4)$

对于标准测试问题, $S(z) = 1 + z + z^2/2! + z^3/3! + z^4/4!$

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:

Stability Region:

$$S(z) = 1 + z + z^2/2! + z^3/3! + z^4/4!$$



• ④ Kutta's 3/8 Rule:

$$s_1 = f_n = f(t_n, u_n)$$

$$s_2 = f\left(t_n + \frac{h}{3}, u_n + \frac{h}{3}s_1\right)$$

$$s_3 = f\left(t_n + \frac{2h}{3}, u_n - \frac{h}{3}s_1 + hs_2\right)$$

$$s_4 = f(t_n + h, u_n + hs_1 - hs_2 + hs_3)$$

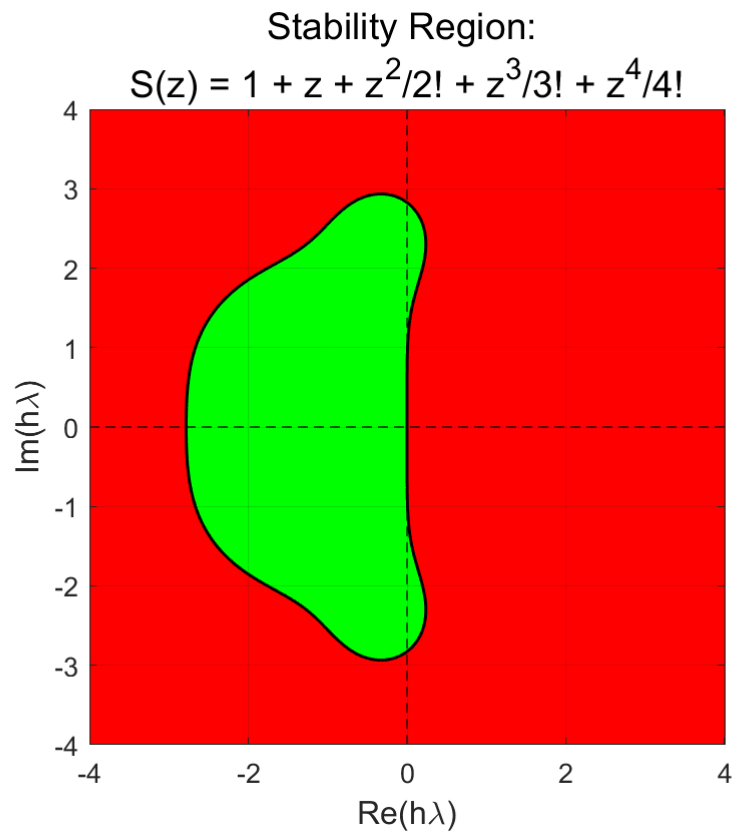
$$u_{n+1} = u_n + \frac{h}{8}(s_1 + 3s_2 + 3s_3 + s_4)$$

c_1	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	$a_{1,4}$	0	0	0	0	0
c_2	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$	$a_{2,4}$	$1/3$	$1/3$	0	0	0
c_3	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$	$a_{3,4}$	$2/3$	$-1/3$	1	0	0
c_4	$a_{4,1}$	$a_{4,2}$	$a_{4,3}$	$a_{4,4}$	1	1	-1	1	0
	b_1	b_2	b_3	b_4		$1/8$	$3/8$	$3/8$	$1/8$

单步截断误差为 $O(h^5)$, 全局截断误差为 $O(h^4)$

对于标准测试问题, $S(z) = 1 + z + z^2/2! + z^3/3! + z^4/4!$

绝对稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 如图所示:



显式 Runge-Kutta 法对于标准测试问题不可能是绝对稳定的.

(即 $\{z \in \mathbb{C} : \operatorname{Re}(z) < 0\}$ 包含在稳定域 $\mathcal{A} = \{z \in \mathbb{C} : |S(z)| < 1\}$ 中)

这是因为 p 阶显式 Runge-Kutta 法的有理函数 $S(z) = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^p}{p!}$ 满足:

$$\lim_{\operatorname{Re}(z) \rightarrow -\infty} |S(z)| = \infty$$

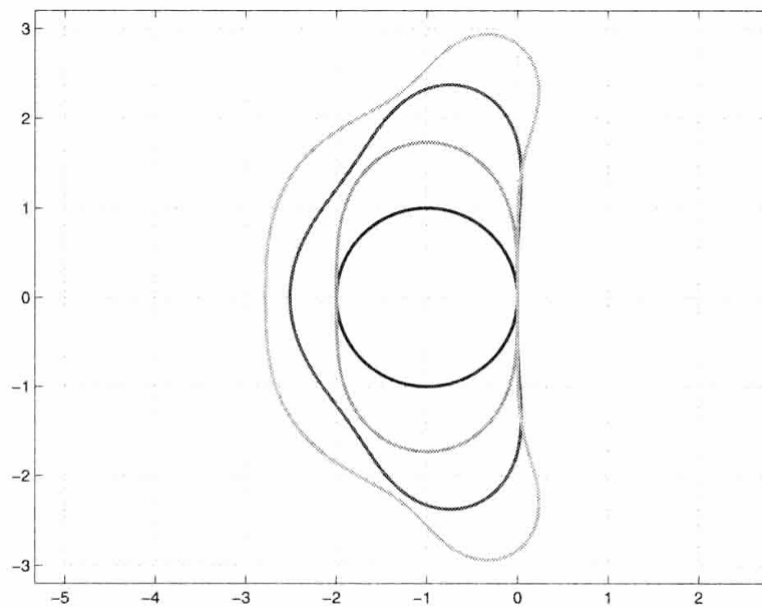


图 2.5: Runge Kutta方法的绝对稳定区域

- Sometimes long-term or asymptotic ($t \rightarrow +\infty$) behavior of an autonomous IVP is of interest
- Model problem ($\lambda \in \mathbb{C}$)

$$\begin{cases} u'(t) = \lambda u(t), & (t > 0) \\ u(0) = u_0 \neq 0 \end{cases}$$

Exact solution: $u(t) = u_0 e^{\lambda t}$

$$\rightsquigarrow \lim_{t \rightarrow +\infty} u(t) = 0 \Leftrightarrow \operatorname{Re}(\lambda) < 0$$

- Runge–Kutta method is of the form $u_k = (S(\lambda h))^k u_0$ for some rational function $S(z)$
- Stability region: $\mathcal{A}(\lambda) = \{z \in \mathbb{C} : |S(z)| < 1\}$
- Absolute stability: $\{z \in \mathbb{C} : \operatorname{Re}(z) < 0\} \subset \mathcal{A}(\lambda)$
 $\rightsquigarrow$ any step size is fine (for stability)
- Examples
 - Forward Euler method: $S(z) = 1 + z$
 - Backward Euler method: $S(z) = (1 - z)^{-1}$
 $\rightsquigarrow$ absolutely stable
 - Trapezoidal method: $S(z) = (1 + z/2)(1 - z/2)^{-1}$
 $\rightsquigarrow$ absolutely stable
 - RK4: $S(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$
- Explicit Runge–Kutta methods cannot be absolutely stable because $S(z)$ is a polynomial

$$\lim_{z \rightarrow -\infty} |S(z)| = +\infty$$

推广标准测试问题的收敛域结论:

- A general autonomous IVP

$$\begin{cases} u'(t) = f(u(t)), & (t > 0) \\ u(0) = u_0 \neq 0 \end{cases}$$

with stationary point u_* :

$$\lim_{t \rightarrow +\infty} u(t) = u_*, \quad (\text{for } u_0 \text{ sufficiently close to } u_*)$$

- Linearization:

$$u'(t) \approx J(u_*)u(t)$$

$\rightsquigarrow$ Runge–Kutta method with step size $h \in \cap_{\lambda \in \text{spec}(J(u_*))} \mathcal{A}(\lambda)$ converges to u_* if u_0 is sufficiently close to u_*

- Implicit methods are preferred for stiff problems

- Another case:

$$\begin{cases} u'(t) = iHu(t), & (t > 0) \\ u(0) = u_0 \neq 0 \end{cases}$$

with $H^* = H$

Exact solution $u(t) = e^{iHt}u_0 \rightsquigarrow \|u(t)\|_2$ is a constant

- Forward Euler method: $\lim_{k \rightarrow \infty} \|u_k\|_2 = +\infty$
- Backward Euler method: $\lim_{k \rightarrow \infty} \|u_k\|_2 = 0$
 $\rightsquigarrow$ too stable!
- Trapezoidal method: $\|u_k\|_2 = \|u_0\|_2$
 h is on the boundary of $\mathcal{A}(\lambda)$

(自适应步长调整)

我们可以使用高阶方法来预估局部截断误差，从而自适应调整低阶方法的步长。
(可参考微分方程数值解第 2.3.3 节)

Adaptive step size

- In practice, a large step size is preferred
- Equispaced nodes might be an overkill for some subintervals
- Adaptive step size: $LTE \approx tol$
- Method:
 - Use the same Runge–Kutta method with a smaller step size
 - Use a higher order Runge–Kutta method with the same step size
↪ MATLAB functions ode23, ode45

其中 tol 是截断误差的阈值.

5.3 有限差分法

有限差分法使用数值微分的有限差分公式来替代原有微分, 将原微分方程转化为大型稀疏线性方程组.

以 Poisson 方程为例:

- Poisson's equation:

$$-\nabla^2 u(x) = f(x), \quad x \in \Omega$$

- Laplace equation: $\nabla^2 u(x) = 0$
- Harmonic functions: solutions of the Laplace equation
- Typical boundary conditions (BC):
 - Dirichlet boundary condition: $u(x) = 0$ for $x \in \partial\Omega$
 - Neumann boundary condition: $\nabla u(x) = 0$ for $x \in \partial\Omega$
 - Robin boundary condition: $au(x) + b\nabla u(x) = 0$ for $x \in \partial\Omega$
 - Mixed boundary condition
- ↪ Under one of these boundary conditions, $\nabla u(x) = 0$ in Ω

5.3.1 一维 Poisson 方程

考虑一维 Poisson 方程:

$$-\frac{d^2}{dx^2}u(x) = f(x) \quad (0 < x < 1)$$

其中初值为 $u(0) = u(1) = 0$ (即 Dirichlet 边界条件)

我们使用有限差分去逼近上述方程:

$$\begin{aligned} -\frac{d^2}{dx^2}u(x) &= f(x) \quad (0 < x < 1) \\ \Updownarrow \\ \frac{2u_i - u_{i-1} - u_{i+1}}{h^2} + O(h^2) &= f_i \quad (i = 1, \dots, n) \\ \text{where } h &:= \frac{1}{n+1}, u_i = u(ih) \text{ and } f_i = f(ih) \end{aligned}$$

我们丢弃截断误差, 将其改写为:

$$\begin{aligned} -u_{i-1} + 2u_i - u_{i+1} &= h^2 f_i \quad (i = 1, \dots, n) \\ \Leftrightarrow \\ T_n u &= \begin{bmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = h^2 \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} = h^2 f \end{aligned}$$

对称正定三对角阵 T_n 的特征值和单位特征向量为:

$$\lambda_k = 2 \left(1 - \cos \left(\frac{k}{n+1} \pi \right) \right) \quad x^{(k)} = \sqrt{\frac{2}{n+1}} \begin{bmatrix} \sin \left(\frac{k\pi}{n+1} \right) \\ \sin \left(\frac{2k\pi}{n+1} \right) \\ \vdots \\ \sin \left(\frac{(n-1)k\pi}{n+1} \right) \\ \sin \left(\frac{nk\pi}{n+1} \right) \end{bmatrix} \quad (k = 1, \dots, n)$$

- 1-D Poisson's equation with Dirichlet BC

$$\begin{cases} -u''(x) = f(x), & (0 < x < 1) \\ u(0) = u(1) = 0 \end{cases}$$

- Finite difference method (FDM) on equispaced grid:

$$\frac{1}{h^2} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n-2} \\ u_{n-1} \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{n-2} \\ f_{n-1} \end{bmatrix}$$

where $h = 1/n$, $x_k = kh$, $u_k \approx u(x_k)$, $f_k = f(x_k)$

- Symmetric positive definite system
- Condition number $= \Theta(h^{-2})$

5.3.2 二维 Poisson 方程

考虑二维 Poisson 方程:

$$-\frac{\partial^2}{\partial x^2} u(x, y) - \frac{\partial^2}{\partial y^2} u(x, y) = f(x, y) \quad (0 < x, y < 1)$$

其中初值为 $\begin{cases} u(x, 0) = u(x, 1) = 0 & (\forall 0 \leq x \leq 1) \\ u(0, y) = u(1, y) = 0 & (\forall 0 \leq y \leq 1) \end{cases}$ (即 Dirichlet 边界条件)

使用有限差分去逼近上述方程可得:

$$\frac{2u_{ij} - u_{i-1,j} - u_{i+1,j}}{h^2} + \frac{2u_{ij} - u_{i,j-1} - u_{i,j+1}}{h^2} = f_{i,j} \quad (1 \leq i, j \leq n)$$

where $h := \frac{1}{n+1}$, $u_{i,j} = u(ih, jh)$ and $f_{i,j} = f(ih, jh)$

$$\Downarrow$$

$$4u_{ij} - u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1} = h^2 f_{i,j} \quad (1 \leq i, j \leq n)$$

$$\Downarrow$$

$$T_n U + U T_n = h^2 F + B$$

$$T_n = \begin{bmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{bmatrix} \quad U = \begin{bmatrix} u_{1,1} & u_{1,2} & \cdots & u_{1,n} \\ u_{2,1} & u_{2,2} & \cdots & u_{2,n} \\ \vdots & \vdots & & \vdots \\ u_{n,1} & u_{n,2} & \cdots & u_{n,n} \end{bmatrix} \quad F = \begin{bmatrix} f_{1,1} & f_{1,2} & \cdots & f_{1,n} \\ f_{2,1} & f_{2,2} & \cdots & f_{2,n} \\ \vdots & \vdots & & \vdots \\ f_{n,1} & f_{n,2} & \cdots & f_{n,n} \end{bmatrix}$$

$$B = \begin{bmatrix} u_{1,0} + u_{0,1} & u_{0,2} & \cdots & u_{0,n-1} & u_{0,n} + u_{1,n+1} \\ u_{2,0} & 0 & \cdots & 0 & u_{2,n+1} \\ \vdots & \vdots & & \vdots & \vdots \\ u_{n-1,0} & 0 & \cdots & 0 & u_{n-1,n+1} \\ u_{n,0} + u_{n+1,1} & u_{n+1,2} & \cdots & u_{n+1,n-1} & u_{n+1,n} + u_{n,n+1} \end{bmatrix}$$

$$\Downarrow$$

$$T_{n \times n} \text{vec}(U) = h^2 \text{vec}(F) + \text{vec}(B)$$

where $T_{n \times n} := (I_n \otimes T_n + T_n \otimes I_n) = \begin{bmatrix} T_n & & & \\ & \ddots & & \\ & & \ddots & \\ & & & T_n \end{bmatrix} + \begin{bmatrix} 2I_n & -I_n & & \\ -I_n & \ddots & \ddots & \\ & \ddots & \ddots & -I_n \\ & & -I_n & 2I_n \end{bmatrix}$

- Finite difference method (FDM) on equispaced mesh:

$$\text{1D: } T_x u = f$$

$$\text{2D: } (I \otimes T_x + T_y \otimes I) u = f$$

$$\text{3D: } (I \otimes I \otimes T_x + I \otimes T_y \otimes I + T_z \otimes I \otimes I) u = f$$

- Symmetric positive definite system

- Condition number: $\Theta((h_x^2 + h_y^2 + h_z^2)^{-1})$

- Frequently used solvers:

- Preconditioned conjugate gradient (PCG) method

- Sparse direct method

- Fast Fourier transform (FFT)

- ...

值得说明的是，我们通常用**多重网格法** (Multigrid, MG) 得到的近似解作为预条件。

多重网格法尽管精度不高，但作为预条件十分合适，

其外层迭代次数很稳定，不随问题规模的增大而显著改变。

5.3.2 其他标准问题

Model problems

- Poisson's equation
- Heat equation:

$$\frac{\partial u(x, t)}{\partial t} = \alpha \nabla^2 u(x, t)$$

with initial condition $u(x, 0) = u_0(x)$

- Wave equation

$$\frac{\partial^2 u(x, t)}{\partial t^2} = \alpha^2 \nabla^2 u(x, t)$$

with initial conditions $u(x, 0) = u_0(x)$, $u_t(x, 0) = 0$

考虑热传导方程:

Finite difference method for parabolic equations

- Heat equation: $u_t(x, t) = \alpha u_{xx}(x, t)$
with initial condition $u(x, 0) = u_0(x)$

- Forward Euler scheme

$$\frac{u_j^{(k+1)} - u_j^{(k)}}{\delta t} = \alpha \frac{u_{j-1}^{(k)} - 2u_j^{(k)} + u_{j+1}^{(k)}}{h^2}$$

- Explicit method
- Truncation error: $O(\delta t + h^2)$
- Stable when $0 < \delta t \leq h^2/(2\alpha)$

- Backward Euler scheme

$$\frac{u_j^{(k+1)} - u_j^{(k)}}{\delta t} = \alpha \frac{u_{j-1}^{(k+1)} - 2u_j^{(k+1)} + u_{j+1}^{(k+1)}}{h^2}$$

- Implicit method
- Truncation error: $O(\delta t + h^2)$
- Stable when $\delta t > 0$

- Crank–Nicolson (CN) scheme

$$\begin{aligned} \frac{u_j^{(k+1)} - u_j^{(k)}}{\delta t} = & \alpha \frac{u_{j-1}^{(k)} - 2u_j^{(k)} + u_{j+1}^{(k)}}{2h^2} \\ & + \alpha \frac{u_{j-1}^{(k+1)} - 2u_j^{(k+1)} + u_{j+1}^{(k+1)}}{2h^2} \end{aligned}$$

- Implicit method
- Truncation error: $O(\delta t^2 + h^2)$
- Stable when $\delta t > 0$

考虑波动方程:

Finite difference method for hyperbolic equations

- Wave equation: $u_{tt}(x, t) = \alpha^2 u_{xx}(x, t)$
with initial conditions $u(x, 0) = u_0(x)$, $u_t(x, 0) = 0$

- Forward difference scheme

$$\frac{u_j^{(k+1)} - 2u_j^{(k)} + u_j^{(k-1)}}{\delta t^2} = \alpha \frac{u_{j-1}^{(k)} - 2u_j^{(k)} + u_{j+1}^{(k)}}{h^2}$$

- Explicit method
- Truncation error: $O(\delta t^2 + h^2)$
- Stable when $0 < \delta t \leq h/\alpha$

- Backward difference and Crank–Nicolson schemes also exist, and are stable when $\delta t > 0$

5.4 有限元方法

5.4.1 投影的思想

考虑无限维空间 Ω 上的算子 A 的特征值问题 $Ax = x\lambda$,

我们可将其投影到有限维子空间上进行求解.

设 V 是无限维空间 Ω 的某个有限维子空间的标准正交基 (例如 Krylov 基)

设 $\text{span}(V)$ 对应的近似特征对为 $(\hat{x}, \hat{\lambda})$

我们要求残差 $r = A\hat{x} - \hat{x}\hat{\lambda}$ 满足 **Galerkin 条件** $r \perp \text{span}(V)$

具体来说, 设 $\hat{x}$ 在 V 下的坐标为 y (即 $\hat{x} = Vy$), 则我们有:

$$\begin{aligned} V^T r &= V^T (A\hat{x} - \hat{x}\hat{\lambda}) \\ &= V^T (AVy - Vy\hat{\lambda}) \\ &= V^T AVy - y\hat{\lambda} \end{aligned}$$

于是 Galerkin 条件 $r \perp \text{span}(V)$ 可翻译为:

$$\begin{aligned} r &\perp \text{span}(V) \\ &\Updownarrow \\ V^T r &= V^T AVy - y\hat{\lambda} = 0_{\dim(V)} \\ &\Updownarrow \\ V^T AVy &= y\hat{\lambda} \end{aligned}$$

因此在给定 V 的条件下, 问题转化为 $\dim(V)$ 维矩阵 $V^T AV$ 的特征值问题.

通过迭代地更新子空间标准正交基 V (例如 Krylov 基的更新方式) 提升解的精度.

我们也可利用投影的思想将无限维常微分方程问题转化为有限维问题.

考虑以下常微分问题:

$$\begin{aligned} -u''(x) &= f(x) \\ u(0) &= u(2\pi) = 0 \end{aligned}$$

其精确解 $u(x)$ 落在无限维空间 $\text{span}\{e^{i2\pi nx} : n = 0, \pm 1, \pm 2, \dots\}$ 中.

现考虑有限维子空间 $\text{span}\{e^{i2\pi nx} : n = 0, \pm 1, \pm 2\}$

记 $u(x)$ 在 $\text{span}\{e^{i2\pi nx} : n = 0, \pm 1, \pm 2\}$ 中的投影为 $\hat{u}(x) = \sum_{k=-2}^2 c_k e^{i2\pi kx}$

它满足 Galerkin 条件:

$$\begin{aligned}\langle \hat{u}''(x) + f(x), e^{i2\pi jx} \rangle &= \left\langle \sum_{k=-2}^2 c_k (e^{i2\pi kx})'' + f(x), e^{i2\pi jx} \right\rangle \\ &= \sum_{k=-2}^2 c_k \cdot (i2\pi k)^2 \langle e^{i2\pi kx}, e^{i2\pi jx} \rangle + \langle f(x), e^{i2\pi jx} \rangle \quad (j = 0, \pm 1, \pm 2) \\ &= -4\pi^2 j^2 c_j + \langle f(x), e^{i2\pi jx} \rangle \\ &= 0\end{aligned}$$

它可写成如下矩阵形式:

$$-4\pi^2 \begin{bmatrix} (-2)^2 & & & & \\ & (-1)^2 & & & \\ & & 0^2 & & \\ & & & 1^2 & \\ & & & & 2^2 \end{bmatrix} \begin{bmatrix} c_{-2} \\ c_{-1} \\ c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} b_{-2} \\ b_{-1} \\ b_0 \\ b_1 \\ b_2 \end{bmatrix}$$

$$\text{where } b_j := \langle f(x), e^{i2\pi jx} \rangle = \int_0^{2\pi} e^{-i2\pi jx} f(x) dx$$

$\Updownarrow$

Can be denoted as $Ac = b$

where $A = [a_{jk}] = [\langle (e^{i2\pi kx})'', e^{i2\pi jx} \rangle]$

Fourier 基的标准正交性保证了系数矩阵 A 是对角阵.

一般来说, 形如 $\alpha u''(x) + \beta u'(x) + \gamma u(x) = f(x)$ 的常系数微分方程

可通过在有限个 Fourier 基函数构成的子空间上进行投影求解, 对应的系数矩阵为三对角阵.

但形如 $\alpha(x)u''(x) + \beta(x)u'(x) + \gamma(x)u(x) = f(x)$ 的变系数微分方程就不一定了,

此时我们需要使用更合适的标准正交基.

考虑以下常微分问题:

$$\begin{aligned}-u''(x) &= f(x) \\ u(a) &= u(b) = 0\end{aligned}$$

其精确解 $u(x)$ 落在无限维空间 $\text{span}\{\phi_n(x) : n = 0, 1, \dots\}$ 中.

根据内积空间的性质可知,

$u''(x) + f(x) \equiv 0$ 等价于对于任意 $\phi \in \text{span}\{\phi_n(x) : n = 0, 1, \dots\}$ 都有 $\langle u''(x) + f(x), \phi \rangle = 0$ 成立.

现考虑有限维子空间 $\text{span}\{\phi_n(x) : n = 0, 1, 2\}$

记 $u(x)$ 在 $\text{span}\{\phi_n(x) : n = 0, 1, 2\}$ 中的投影为 $\hat{u}(x) = \sum_{k=0}^2 c_k \phi_k(x)$

它满足 Galerkin 条件:

$$\begin{aligned}\langle \hat{u}''(x) + f(x), \phi_j(x) \rangle &= \left\langle \sum_{k=0}^2 c_k \phi_k''(x) + f(x), \phi_j(x) \right\rangle \\ &= \sum_{k=0}^2 c_k \langle \phi_k''(x), \phi_j(x) \rangle + \langle f(x), \phi_j(x) \rangle \quad (j = 0, 1, 2) \\ &= 0\end{aligned}$$

它可写成如下矩阵形式:

$$\begin{bmatrix} \langle \phi_0''(x), \phi_0(x) \rangle & \langle \phi_1''(x), \phi_0(x) \rangle & \langle \phi_2''(x), \phi_0(x) \rangle \\ \langle \phi_0''(x), \phi_1(x) \rangle & \langle \phi_1''(x), \phi_1(x) \rangle & \langle \phi_2''(x), \phi_1(x) \rangle \\ \langle \phi_0''(x), \phi_2(x) \rangle & \langle \phi_1''(x), \phi_2(x) \rangle & \langle \phi_2''(x), \phi_2(x) \rangle \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} \langle f(x), \phi_0(x) \rangle \\ \langle f(x), \phi_1(x) \rangle \\ \langle f(x), \phi_2(x) \rangle \end{bmatrix}$$

Can be denoted as $Ac = b$

$$\text{where } A = [a_{jk}] = [\langle \phi_k''(x), \phi_j(x) \rangle] \text{ and } b_j = \langle f(x), \phi_j(x) \rangle = \int_a^b \overline{\phi_j(x)} f(x) dx$$

我们需要尽量让系数矩阵 A 稀疏,

一种方法是使用短程的基函数 (与之相对, Fourier 基是长程的),

让它们的支撑集相互错开, 这样它们自然正交, 使得系数矩阵有更多 0.

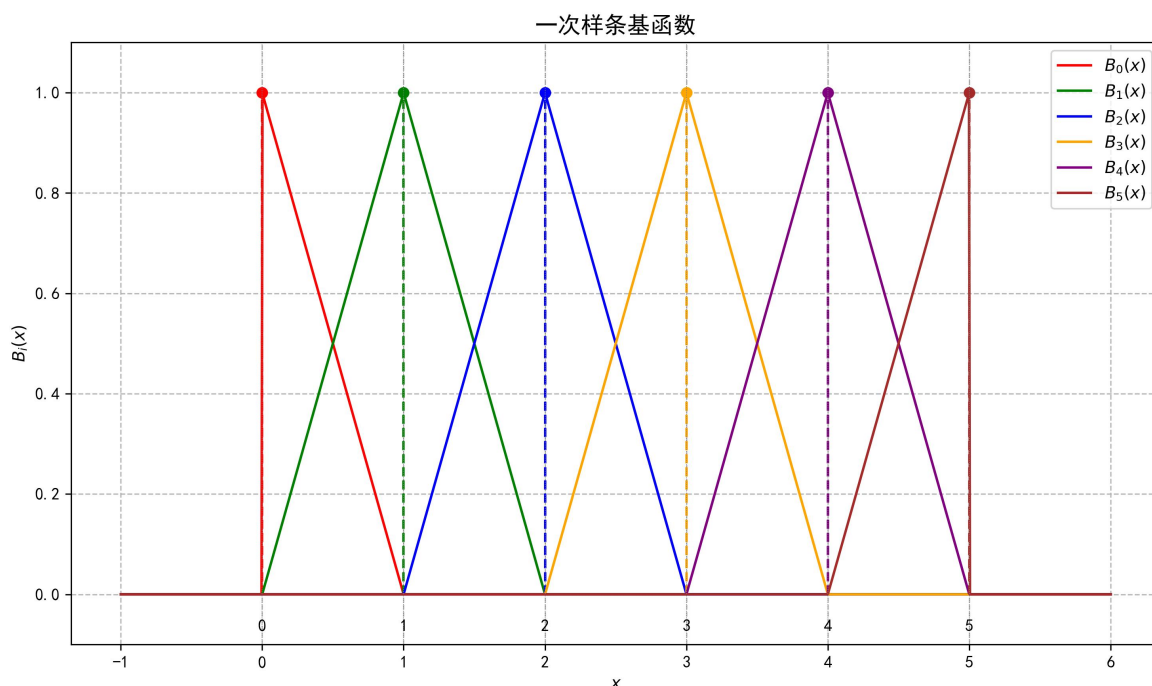
简单起见, 考虑一次样条基函数.

取等距节点 $a = x_0 < x_1 < \dots < x_n = b$, 其中 $x_i = x_0 + ih$, $h = (b - a)/n$

则一次样条基函数的形式为:

$$\begin{aligned} \phi_0(x) &= \begin{cases} \frac{x_1 - x}{h}, & x \in [x_0, x_1] \\ 0, & \text{otherwise} \end{cases} \\ \phi_i(x) &= \begin{cases} \frac{x - x_{i-1}}{h}, & x \in [x_{i-1}, x_i] \\ \frac{x_{i+1} - x}{h}, & x \in (x_i, x_{i+1}] \\ 0, & \text{otherwise} \end{cases} \quad (i = 1, \dots, n-1) \\ \phi_n(x) &= \begin{cases} \frac{x - x_{n-1}}{h}, & x \in [x_{n-1}, x_n] \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

以 $h = 1$, $x_i = i$, $n = 5$ 的情况为例:



5.4.2 具体形式

一次样条基函数非常简单, 可问题是它在斜率变化点处的导数不存在 (更不用说二阶导数了)

为此我们定义**广义导数**.

设 $C_0^\infty[a, b]$ 是 (a, b) 上无限光滑, 且具有包含于 (a, b) 的紧支撑集的函数构成的集合.

给定函数 v (不一定可导), 对于任意 $\phi \in C_0^\infty[a, b]$, 我们都有:

$$\begin{aligned}\int_a^b v'(x)\phi(x)dx &= v(x)\phi(x)\Big|_a^b - \int_a^b v(x)\phi'(x)dx \quad (\text{note that } \phi(a) = \phi(b) = 0) \\ &= - \int_a^b v(x)\phi'(x)dx\end{aligned}$$

- Weak derivative of $v(x)$

$$\int_0^1 v'(x)\phi(x)dx = - \int_0^1 v(x)\phi'(x)dx$$

for all $\phi(x) \in C_0^\infty[0, 1]$

- Weak formulation of 1-D Poisson's equation

$$\begin{aligned}- \int_0^1 u''(x)v(x)dx &= \int_0^1 f(x)v(x)dx \\ \Leftrightarrow \int_0^1 u'(x)v'(x)dx &= \int_0^1 f(x)v(x)dx\end{aligned}$$

for all $v(x) \in C_0[0, 1]$

All derivatives are weak ones

- Weak formulation for 2-D/3-D Poisson's equation

$$\int_{\Omega} (\nabla u(x))^T (\nabla v(x)) dx = \int_{\Omega} f(x)v(x) dx$$

for all $v(x) \in C_0(\Omega)$

- Discretized form $Ac = b$, where

$$A_{ij} = \int_{\Omega} (\nabla v_i(x))^T (\nabla v_j(x)) dx, \quad b_i = \int_{\Omega} f(x)v_i(x) dx$$

$\rightsquigarrow$ symmetric positive definite system

- Generalizable to
 - More general differential operators
 - More complicated domain
 - Higher degree polynomials or other basis

对于一次样条基函数来说, 我们有:

$$\begin{aligned}a_{jk} &= \langle \phi_k''(x), \phi_j(x) \rangle \\ &= \langle \phi_k'(x), \phi_j'(x) \rangle\end{aligned}$$

由于只有相邻的基函数的支撑集才相交, 故当且仅当 $|j - k| \leq 1$ 时 a_{jk} 才是非零的.

换言之, 系数矩阵 A 是一个三对角阵.

更具体地说, 系数矩阵 A 是对称三对角阵,

因为其元素具有 (共轭) 对称性: $a_{jk} = \langle \phi_k'(x), \phi_j'(x) \rangle = \langle \phi_j'(x), \phi_k'(x) \rangle = a_{kj}$.

- 考虑对角元 (即 $j = k$ 的情况):

$$\begin{aligned}
a_{jj} &= \langle \phi'_j(x), \phi'_j(x) \rangle \\
&= \begin{cases} \int_{x_0}^{x_1} \left(-\frac{1}{h}\right)^2 dx, & j = 0 \\ \int_{x_{j-1}}^{x_j} \left(\frac{1}{h}\right)^2 dx + \int_{x_j}^{x_{j+1}} \left(-\frac{1}{h}\right)^2 dx, & j = 1, 2, \dots, n-1 \\ \int_{x_{n-1}}^{x_n} \left(\frac{1}{h}\right)^2 dx, & j = n \end{cases} \\
&= \begin{cases} \frac{1}{h}, & j = 0 \\ \frac{2}{h}, & j = 1, 2, \dots, n-1 \\ \frac{1}{h}, & j = n \end{cases}
\end{aligned}$$

- 考虑次对角元和超对角元 (即 $|j - k| = 1$ 的情况):

$$\begin{aligned}
a_{j,j+1} &= \langle \phi'_{j+1}(x), \phi'_j(x) \rangle \\
&= \int_{x_j}^{x_{j+1}} \frac{1}{h} \cdot \left(-\frac{1}{h}\right) dx \\
&= -\frac{1}{h^2} \cdot h \\
&= -\frac{1}{h}
\end{aligned}$$

Since A is symmetric, we have $a_{j+1,j} = a_{j,j+1} = -\frac{1}{h}$

因此系数矩阵具有如下形式:

$$\begin{aligned}
A &= \begin{bmatrix} \langle \phi''_0(x), \phi_0(x) \rangle & \langle \phi''_1(x), \phi_0(x) \rangle & \cdots & \langle \phi''_{n-1}(x), \phi_0(x) \rangle & \langle \phi''_n(x), \phi_0(x) \rangle \\ \langle \phi''_0(x), \phi_1(x) \rangle & \langle \phi''_1(x), \phi_1(x) \rangle & \cdots & \langle \phi''_{n-1}(x), \phi_1(x) \rangle & \langle \phi''_n(x), \phi_1(x) \rangle \\ \vdots & \vdots & & \vdots & \\ \langle \phi''_0(x), \phi_{n-1}(x) \rangle & \langle \phi''_1(x), \phi_{n-1}(x) \rangle & \cdots & \langle \phi''_{n-1}(x), \phi_{n-1}(x) \rangle & \langle \phi''_n(x), \phi_{n-1}(x) \rangle \\ \langle \phi''_0(x), \phi_n(x) \rangle & \langle \phi''_1(x), \phi_n(x) \rangle & \cdots & \langle \phi''_{n-1}(x), \phi_n(x) \rangle & \langle \phi''_n(x), \phi_n(x) \rangle \end{bmatrix} \\
&= \frac{1}{h} \begin{bmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & \ddots & \ddots & \\ & & \ddots & 2 & -1 \\ & & & -1 & 1 \end{bmatrix}
\end{aligned}$$

右端项 b 的元素 $b_j = \langle f(x), \phi_j(x) \rangle = \int_a^b \phi_j(x) f(x) dx$ 如无解析表达式, 则可通过数值积分方式逼近.

值得注意的是, $x_0, x_1, \dots, x_n$ 并不一定要是等距节点,

不过形式会更复杂一些 (参考 Homework 13 Problem 1 的代码实现).

Finite element method (FEM)—using linear element

- Let $\mathcal{V}_h$ be the subspace of $C_0[0, 1]$ that consists of piecewise linear functions with respect to a given grid $0 = x_0 < x_1 < \dots < x_n = 1$
- $v_k(x)$'s are the basis of $\mathcal{V}_h$, $k \in \{1, 2, \dots, n-1\}$
- Approximate $u(x)$ by functions in $\mathcal{V}_h$:

$$u(x) \approx \sum_{k=1}^{n-1} c_k v_k(x)$$

- The discretized linear system is of the form $Ac = b$, where

$$A_{ij} = \int_0^1 v_i'(x) v_j'(x) dx, \quad b_i = \int_0^1 f(x) v_i(x) dx$$

$\rightsquigarrow$ symmetric positive definite system

Finite element method (FEM)

- Idea
 - Transform to weak formulation
 - Project to a finite dimensional subspace (Galerkin method)
 - Subdivide the system into smaller parts (finite elements)
- Advantages over FDM
 - Easier to handle irregular domain
 - Better approximation quality
 - More sound mathematical foundation
- Disadvantage compared to FDM: more complicated

The End